EE 357- Communication Systems

Laboratory Session 2 - Frequency Modulation

OBJECTIVE : Investigate Frequency Modulation

APPARATUS : Mixed domain oscilloscope (Tektronix MDO3014)
Function generator (LG 1301) - (for VCO)
Function generator (AFG 2005)
DC Power supply

THEORY:

Modulation

FM signal can be mathematically represented as,

$$X_{FM}(t) = A_c \cos\left(\omega_c t + 2\pi k_f \int x_m(t)dt\right) \quad \text{.....(1)}$$

For single tone modulation,

$$\begin{aligned} X_{FM}(t) &= A_c \cos\left(\omega_c t + 2\pi\, k_f \int A_m \cos(\omega_m t)\, dt\right) \\ &= A_c \cos\left(\omega_c t + 2\pi\, \Delta f \int \cos(\omega_m t)\, dt\right) \\ &= A_c \cos\left(\omega c\, t + \beta \sin(\omega_m t)\right) \end{aligned} \quad \text{.....(2)}$$

Where;

K_f - Sensitivity of the function generator
$x_m(t)$ - Modulating signal
Δf - Peak frequency deviation
β – Modulation Index

Using the Bessel function it can be shown that,

$$X_{FM}(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos(\omega_c + n\omega_m)\, t \quad \text{.....(3)}$$

And

$$X_{FM}(f) = \frac{A_c}{2} \sum_{n=-\infty}^{\infty} J_n(\beta)\{\delta[f + (f_c + nf_m)] + \delta[f - (f_c + nf_m)]\} \quad \text{.....(4)}$$

Total average power

Total average power of FM signal is given by

$$P_{av} = S_T = \sum_{n=-\infty}^{\infty} \frac{1}{2} A_c^2 J_n^2(\beta) \quad \text{.....(5)}$$

Band width (BW)

Instantaneous frequency of FM signal is given by

$f_i = f_c + k_f\, m(t)$.....(6)

According to Carson's rule, bandwidth of a FM signal is given by

BW = 2 fm (β + 1) ..... (7)

Frequency spectrum of the FM signal which correspond to a specific message signal can be changed by changing β value. As $\beta = k_f A_m/f_m$, by changing A_m, frequency spectrum of the FM signal is changed.

Sensitivity of the function generator

In this experiment, we use a function generator as a voltage controlled oscillator to generate a FM signal as shown in figure 1.

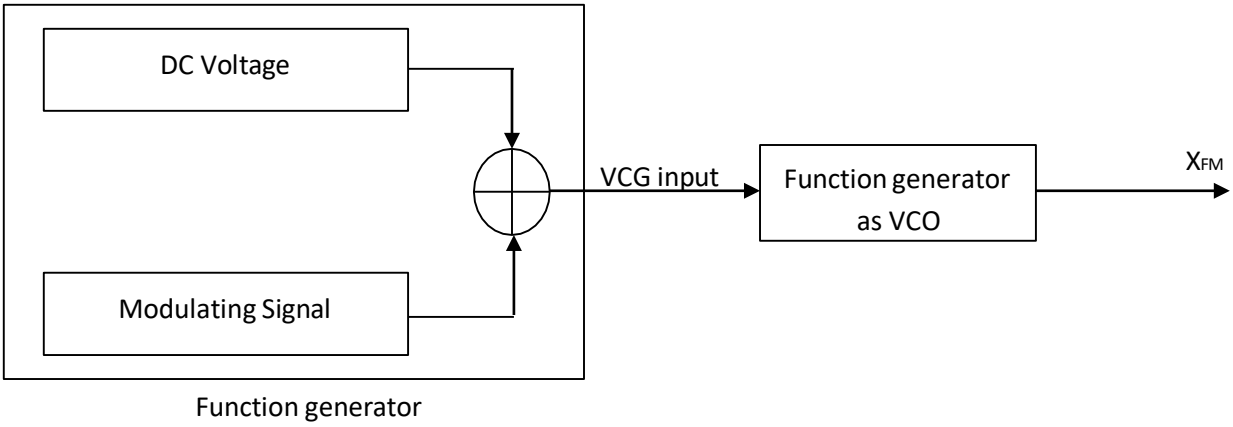

Figure 1: Modulator

Sensitivity of the function generator is a function generator specific parameter. Thus k_f has to be calculated before doing meaningful FM work using the function generator.

k_f can be calculated by feeding known DC signals as VCO inputs and then by analyzing the VCO output signal frequency.

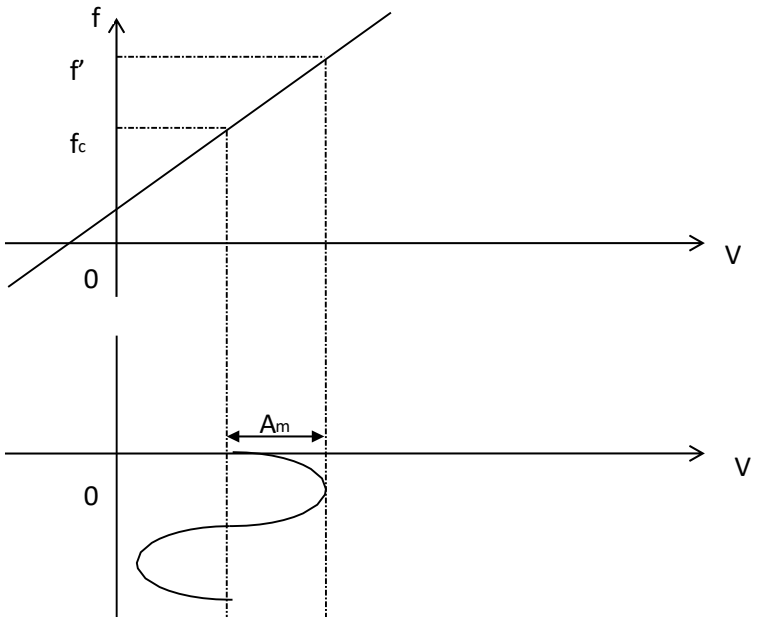

Figure 2: Variation frequency with DC voltage

The VCO output signal frequency can be written as,

$$f_{VCO} = f_c + k_f V_{DC}$$

Therefore, using the gradient of the plot (f_{VCO} vs V_{DC}), the sensitivity can be calculated.

PROCEDURE :

Part 1 : Identify the Sensitivity of the function generator

1. Connect the circuit as shown in figure 1 with no modulation signal applied.
2. Adjust the free running frequency of the VCO to be 1 MHz and its output amplitude to be 400mV while maintaining the V_{DC} at 0 V.
3. Change the V_{DC} voltage level in steps of 0.5 V from -3V to +5V and observe the variation of carrier frequency.
4. Check whether the sensitivity is changing linearly with respect to voltage.
5. Find the sensitivity of the VCO using the observations.

Part 2 : Generate different frequency modulated signal which has different frequency spectrums for the same message signal

1. Connect the circuit as shown in figure 1 with no DC voltage applied.
2. Connect the output of the VCO to mixed domain oscilloscope time domain and frequency spectrum ports.
3. Select frequency of the modulating signal (f_m) to be about 10 kHz.

4. Calculate the desired frequency deviation values that has to be set in order to get the following FM signal frequency spectrum cases
   a) Carrier only.
   b) Only the carrier and the 1st sideband pair are there.
   c) Only the carrier, 1st sideband pair and the 2nd sideband pair are there.
   d) Only the carrier, 1st sideband pair the 2nd sideband pair and 3rd sideband pair are there.
   e) Suppressed carrier.
   f) Suppressed 1st sideband pair.
5. Then by observing the frequency spectrum of the FM signal using the spectrum analyzer, Obtain the above cases by varying frequency deviation.
6. Record frequency deviation value in each case to calculate the modulation index.
7. Compare the previously calculated β value with the β value calculated using the observations.

<u>Part 3 : Observe the effect of fm on the frequency spectrum</u>

For the case of β=4, change f_m and observe the frequency spectrum of the FM signals.

<u>DISCUSSION</u> :

1. Explain why FM is better than AM in terms of resistance to noise.
2. Describe two methods to estimate the occupied bandwidth of an FM signal.
3. Describe the key features of spectrum allocation used for typical FM radio broadcasting.
4. Compare the performance of FM and AM on the frequency spectrum.

Bessel table

Beta	J(0)	J(1)	J(2)	J(3)	J(4)	J(5)	J(6)	J(7)	J(8)	J(9)	J(10)	J(11)	J(12)	J(13)	J(14)
0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.25	0.9844	0.124	0	0	0	0	0	0	0	0	0	0	0	0	0
0.5	0.9385	0.2423	0.0306	0	0	0	0	0	0	0	0	0	0	0	0
0.75	0.8642	0.3492	0.0671	0	0	0	0	0	0	0	0	0	0	0	0
1	0.7652	0.4401	0.1149	0.0196	0	0	0	0	0	0	0	0	0	0	0
1.25	0.6459	0.5106	0.1711	0.0369	0	0	0	0	0	0	0	0	0	0	0
1.5	0.5118	0.5579	0.2321	0.061	0.0118	0	0	0	0	0	0	0	0	0	0
1.75	0.369	0.5802	0.294	0.0919	0.0209	0	0	0	0	0	0	0	0	0	0
2	0.2239	0.5767	0.3528	0.1289	0.034	0	0	0	0	0	0	0	0	0	0
2.25	0.0827	0.5484	0.4047	0.1711	0.0515	0.0121	0	0	0	0	0	0	0	0	0
2.4	0.0025	0.5202	0.431	0.1981	0.0643	0.0162	0	0	0	0	0	0	0	0	0
2.5	-0.0484	0.4971	0.4461	0.2166	0.0738	0.0195	0	0	0	0	0	0	0	0	
2.75	-0.1641	0.426	0.4739	0.2634	0.1007	0.0297	0	0	0	0	0	0	0	0	
3	-0.2601	0.3391	0.4861	0.3091	0.132	0.043	0.0114	0	0	0	0	0	0	0	0
3.5	-0.3801	0.1374	0.4586	0.3868	0.2044	0.0804	0.0254	0	0	0	0	0	0	0	0
4	-0.3971	-0.066	0.3641	0.4302	0.2811	0.1321	0.0491	0.0152	0	0	0	0	0	0	0
4.5	-0.3205	-0.2311	0.2178	0.4247	0.3484	0.1947	0.0843	0.03	0.0091	0	0	0	0	0	0
4.75	-0.2551	-0.2892	0.1334	0.4015	0.3738	0.228	0.1063	0.0405	0.0131	0	0	0	0	0	0
5	-0.1776	-0.3276	0.0466	0.3648	0.3912	0.2611	0.131	0.0534	0.0184	0	0	0	0	0	0
5.5	-0.0068	-0.3414	-0.1173	0.2561	0.3967	0.3209	0.1868	0.0866	0.0337	0.0113	0	0	0	0	0
6	0.1506	-0.2767	-0.2429	0.1148	0.3576	0.3621	0.2458	0.1296	0.0565	0.0212	0	0	0	0	0
6.5	0.2601	-0.1538	-0.3074	-0.0353	0.2748	0.3736	0.2999	0.1801	0.088	0.0366	0.0133	0	0	0	0
7	0.3001	-0.0047	-0.3014	-0.1676	0.1578	0.3479	0.3392	0.2336	0.128	0.0589	0.0235	0	0	0	0
7.5	0.2663	0.1352	-0.2303	-0.2581	0.0238	0.2835	0.3541	0.2832	0.1744	0.0889	0.039	0.0151	0	0	0
8	0.1717	0.2346	-0.113	-0.2911	-0.1054	0.1858	0.3376	0.3206	0.2235	0.1263	0.0608	0.0256	0.0096	0	0
8.5	0.0419	0.2731	0.0223	-0.2626	-0.2077	0.0671	0.2867	0.3376	0.2694	0.1694	0.0894	0.041	0.0167	0	0
9	-0.0903	0.2453	0.1448	-0.1809	-0.2655	-0.055	0.2043	0.3275	0.3051	0.2149	0.1247	0.0622	0.0274	0.0108	0
9.5	-0.1939	0.1613	0.2279	-0.0653	-0.2691	-0.1613	0.0993	0.2868	0.3233	0.2577	0.165	0.0897	0.0427	0.0182	0
10	-0.2459	0.0435	0.2546	0.0584	-0.2196	-0.2341	-0.0145	0.2167	0.3179	0.2919	0.2075	0.1231	0.0634	0.029	0.012
10.5	-0.2366	-0.0789	0.2216	0.1633	-0.1283	-0.2611	-0.1203	0.1236	0.2851	0.3108	0.2477	0.1611	0.0898	0.0441	0.0195
11	-0.1712	-0.1768	0.139	0.2273	-0.015	-0.2383	-0.2016	0.0184	0.225	0.3089	0.2804	0.201	0.1216	0.0643	0.0304
11.5	-0.0677	-0.2284	0.0279	0.2381	0.0963	-0.1711	-0.2451	-0.0846	0.1421	0.2823	0.2998	0.239	0.1575	0.0897	0.0454
12	0.0477	-0.2234	-0.0849	0.1951	0.1825	-0.0735	-0.2437	-0.1703	0.0451	0.2304	0.3005	0.2704	0.1953	0.1201	0.065
12.5	0.1469	-0.1655	-0.1734	0.11	0.2262	0.0347	-0.1984	-0.2252	-0.0538	0.1563	0.2789	0.2899	0.2314	0.1543	0.0896

OBSERVATIONS

Part 1: Identify the Sensitivity of the function generator

Table 01: Frequency Vs V_{DC}

V_{DC} / V	Frequency / MHz
-3	0.9718
-2.5	0.9765
-2	0.9812
-1.5	0.9859
-1	0.9906
-0.5	0.9953
0	1.0000
0.5	1.0046
1	1.0097
1.5	1.0144
2	1.0191
2.5	1.0238
3	1.0285

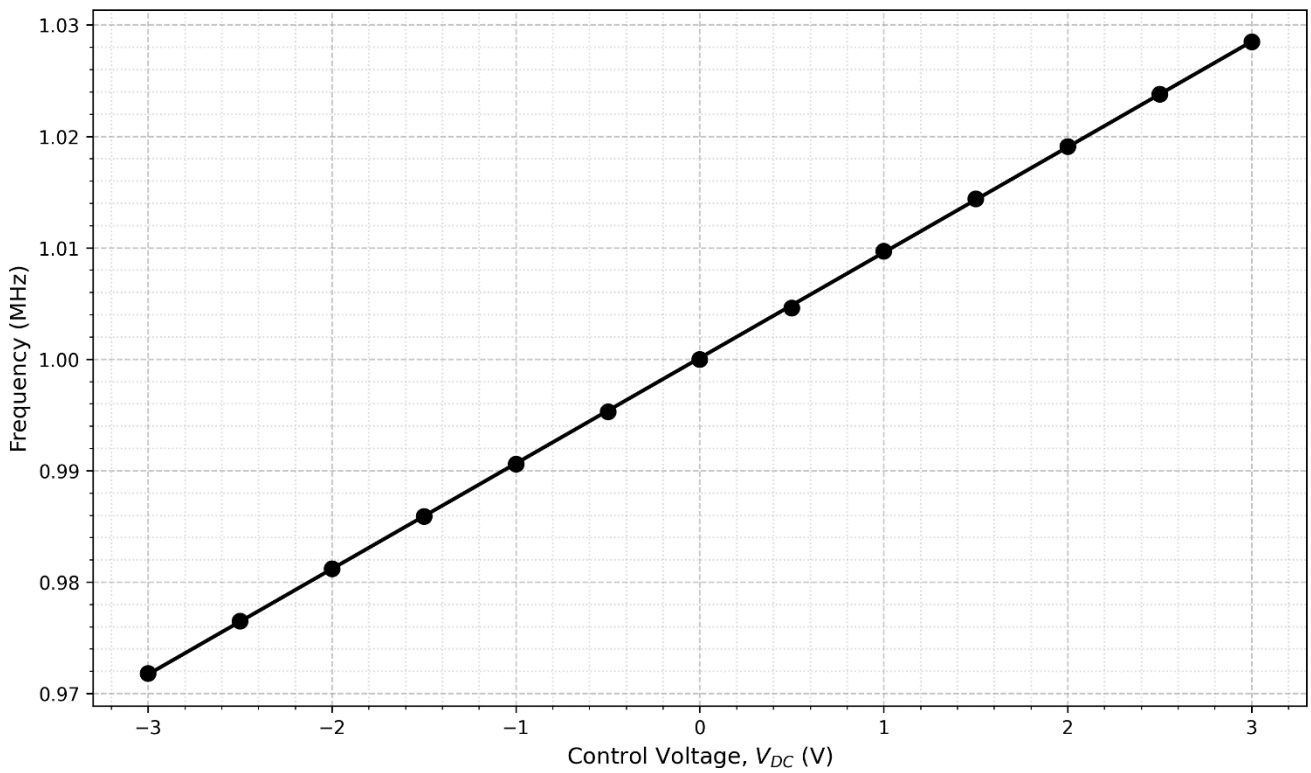

Figure 3 : Frequency Vs V_{DC}

Part 2: Generation of an FM signal

Observe the FM signal through the MDO in time and frequency domains.

a) In frequency domain

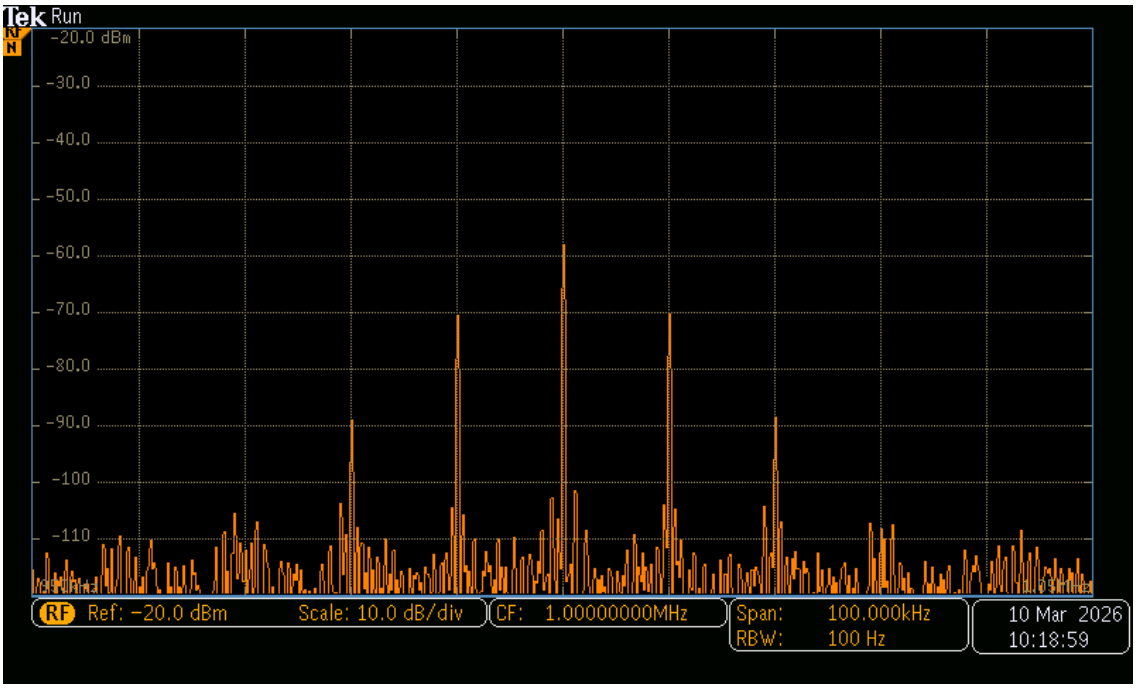

Figure 4: Frequency Spectrum of FM wave

b) In time domain

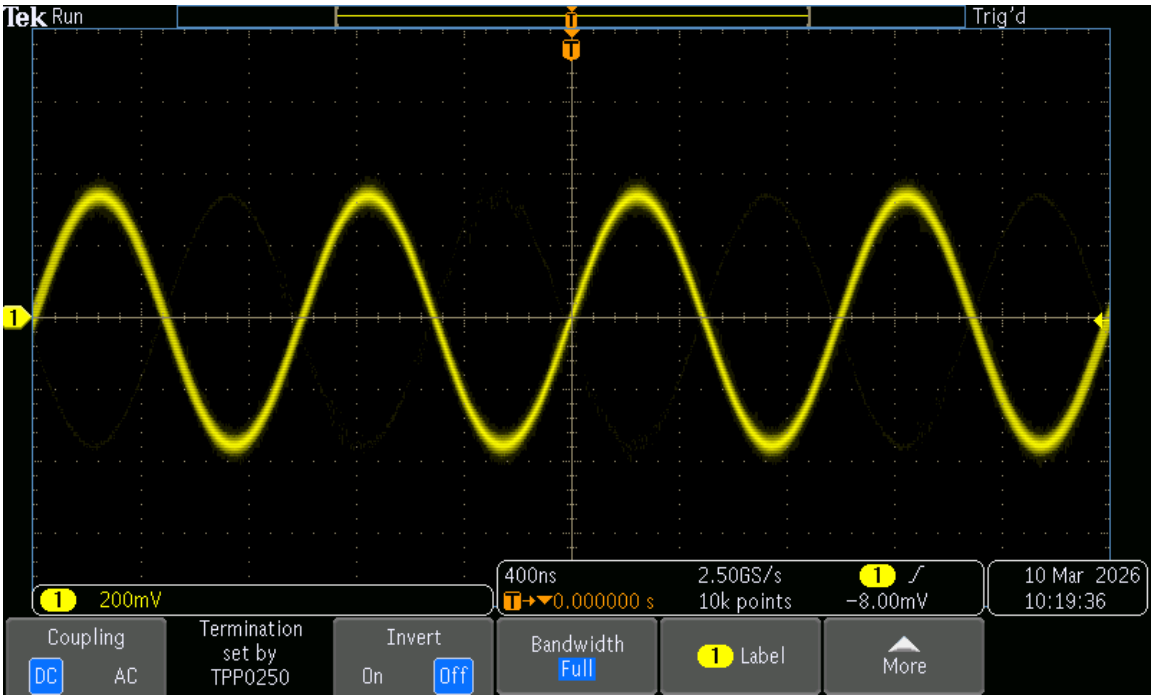

Figure 5: FM signal in time domain

Part 3: Generate different frequency modulated signal which has different frequency spectrums for the same message signal

2)

a) Carrier only

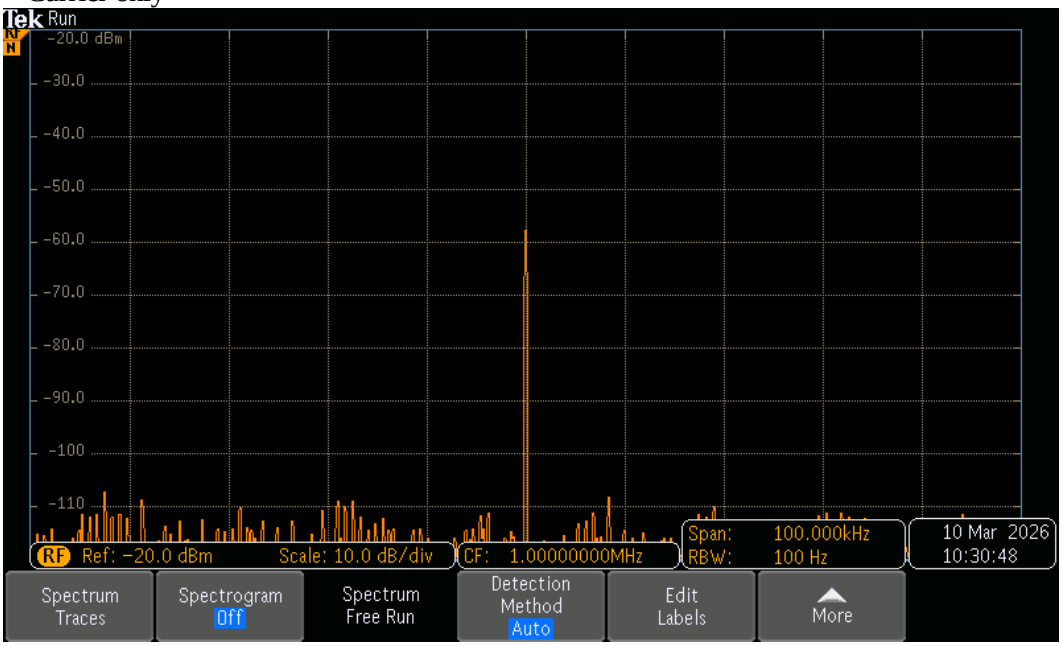

Figure 5: Frequency Spectrum when Modulation Index $\beta = 0$

b) Only the carrier and the 1st sideband pair are there

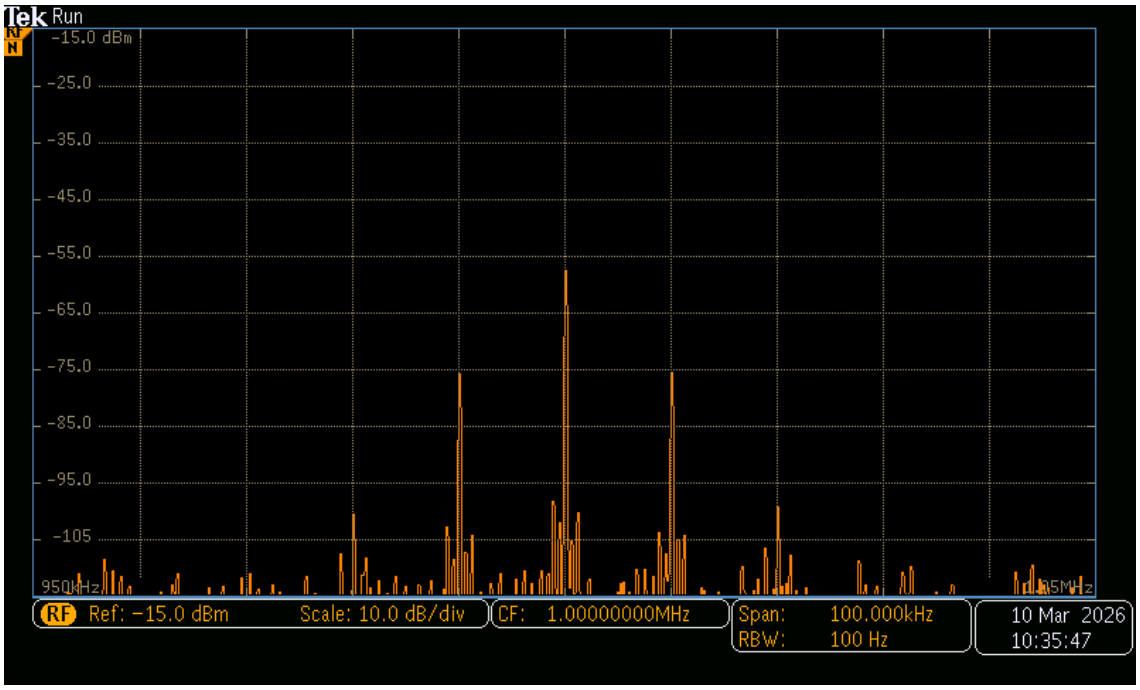

Figure 6: Frequency Spectrum when Modulation Index $\beta = 0.25$

c) Only the carrier, 1st sideband pair and the 2nd sideband pair are there

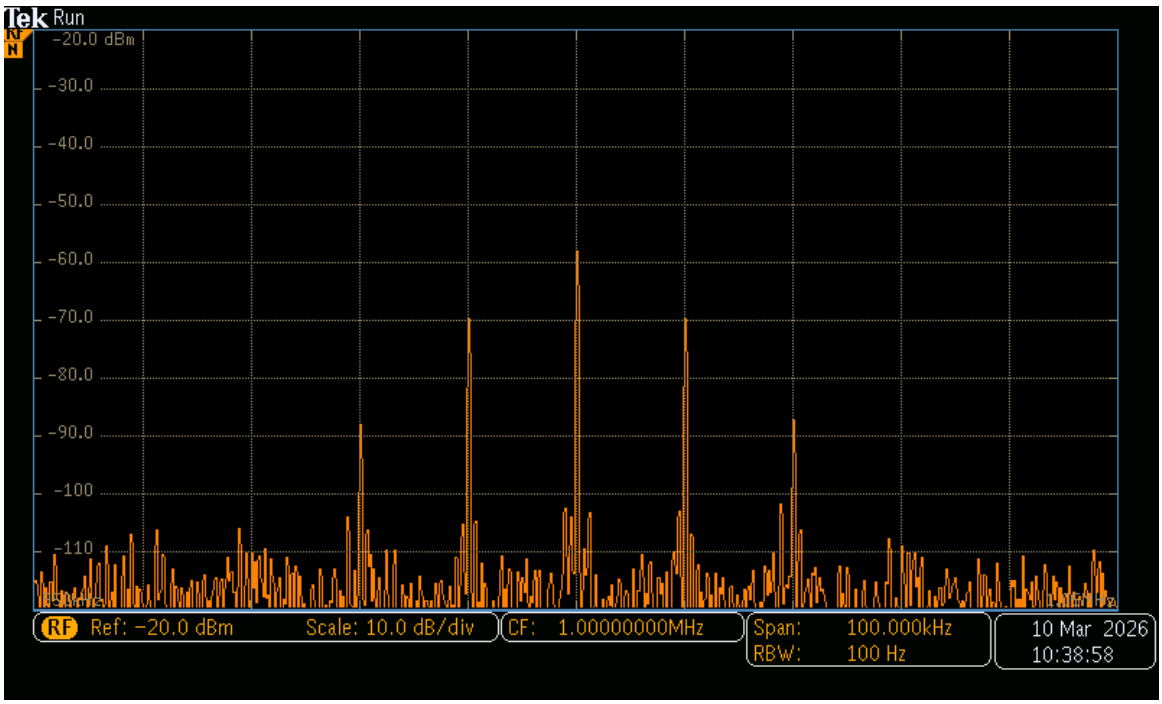

Figure 7: Frequency Spectrum when Modulation Index $\beta = 0.5$

d) Only the carrier, 1st sideband pair, 2nd sideband pair and 3rd sideband pair are there

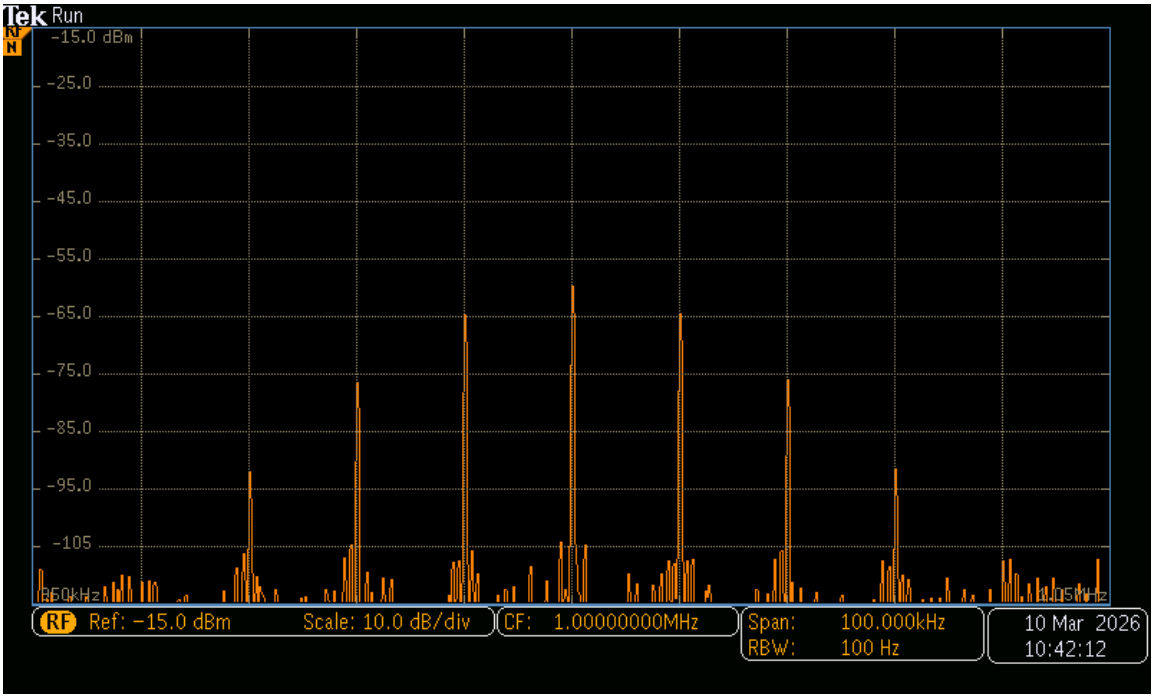

Figure 8: Frequency Spectrum when Modulation Index $\beta = 1$

e) Suppressed carrier

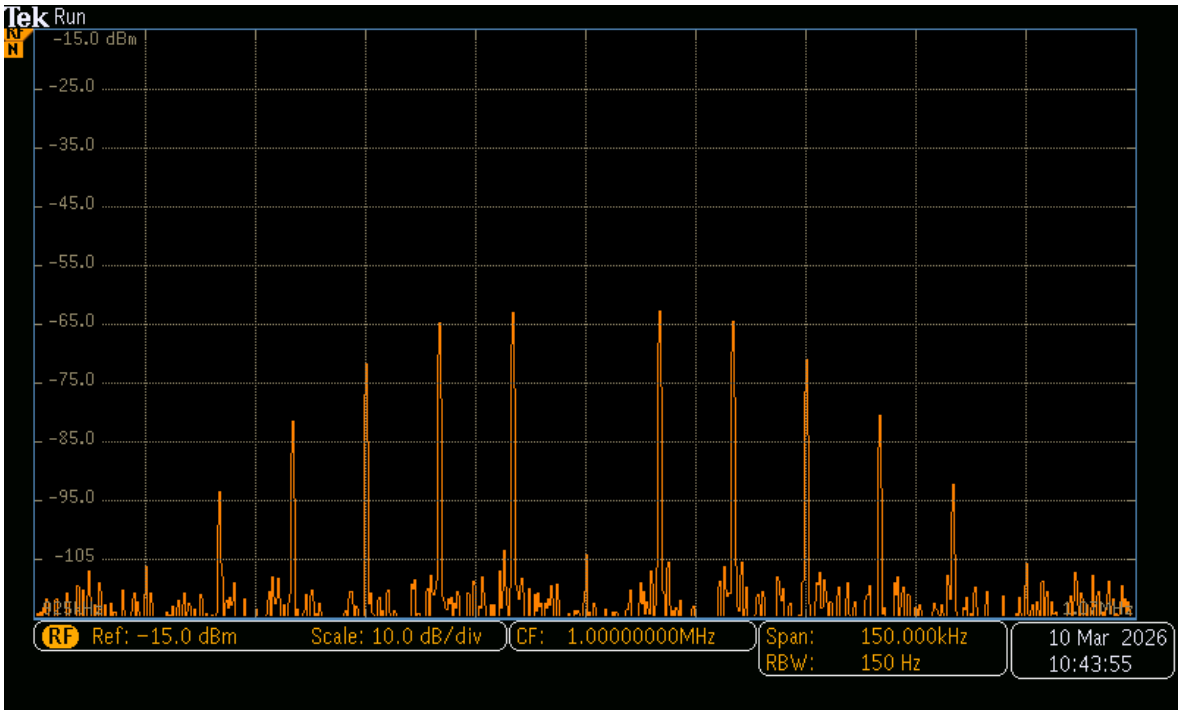

Figure 9: Frequency Spectrum when Modulation Index β = 2.4048

f) Suppressed 1st sideband pair

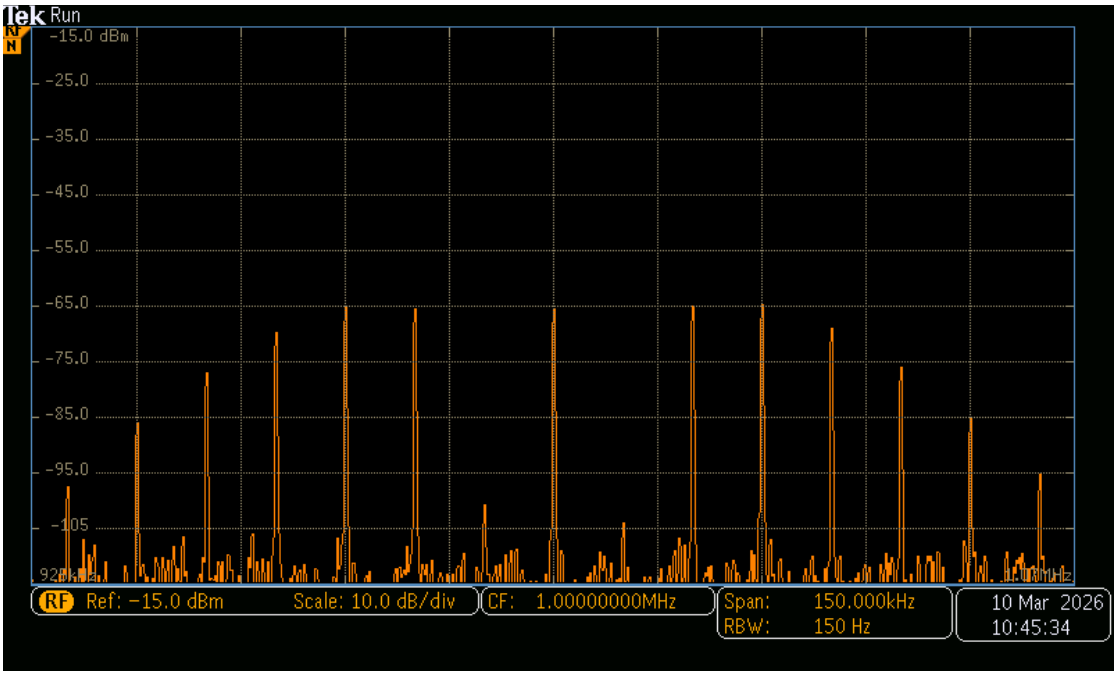

Figure 10: Frequency Spectrum when Modulation Index β = 3.8317

g) Suppressed 2nd sideband pair

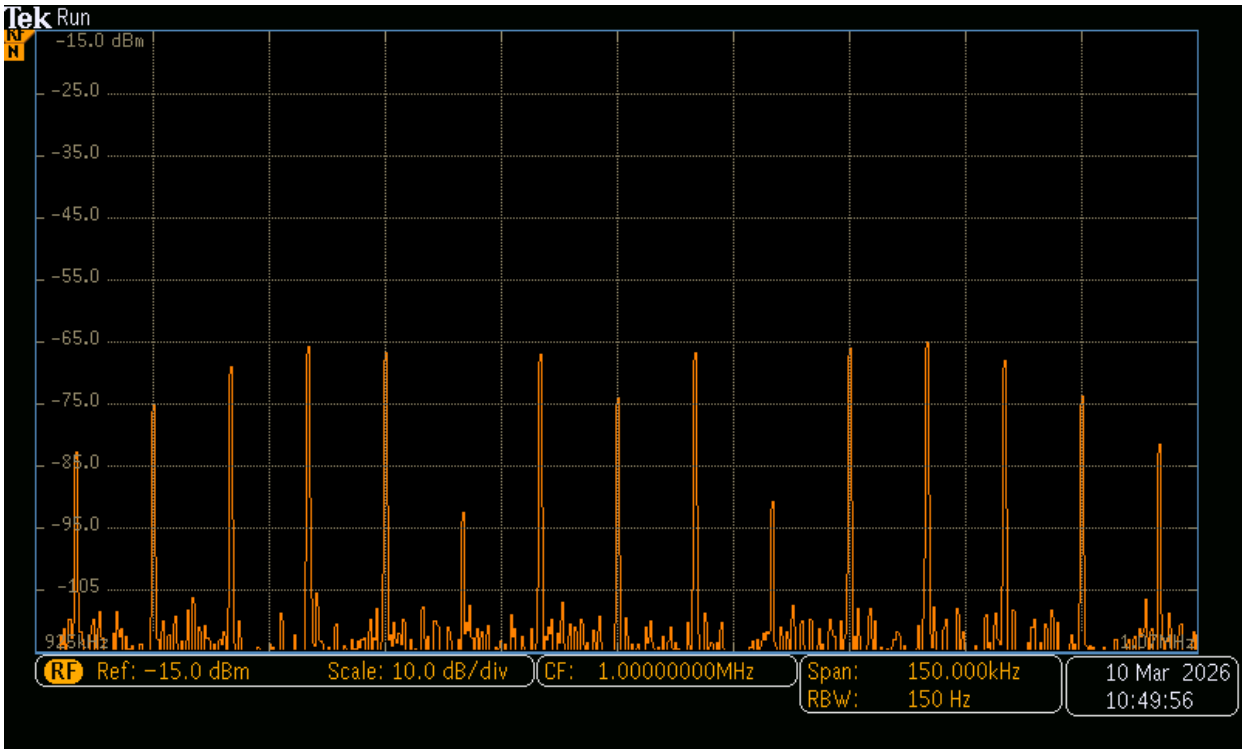

Figure 11: Frequency Spectrum when Modulation Index β = 5.14

Record the practical values of Am' and calculate the modulation index β' for each case.

Table 2: Recorded β values obtained using the Bessel Table and using the Observations of Frequency Spectrum

beta	Am Vp (mv)	Am Vpp (mv)	Am' Vp(mv)	Am' Vpp(mv)
0	0	0	0	0
0.25	264.2706	528.5412	150	300
0.5	528.5412	1057.082	478.5	957
1	1057.082	2114.165	1057	2114
2.4	2536.998	5073.996	2543.5	5087
3.8	4016.913	8033.827	4016.5	8033
5.1	5391.121	10782.24	5384.5	10769

a) Carrier only

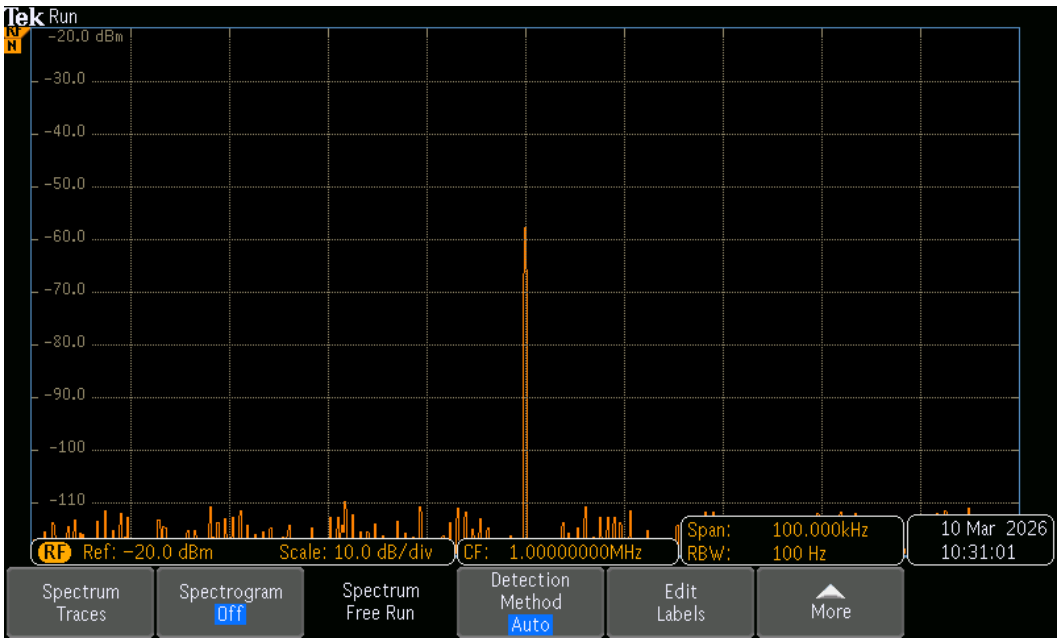

Figure 12: Frequency Spectrum with Carrier

b) Only the carrier and the 1st sideband pair are there

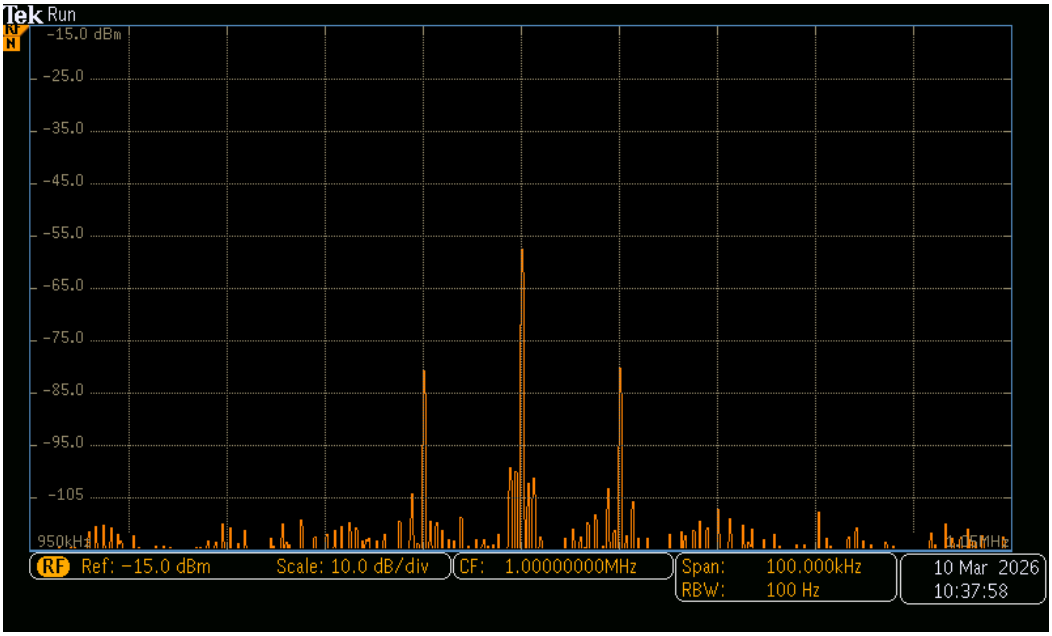

Figure 13: Frequency Spectrum with only the Carrier and First Sideband Pair

c) Only the carrier, 1st sideband pair and the 2nd sideband pair are there

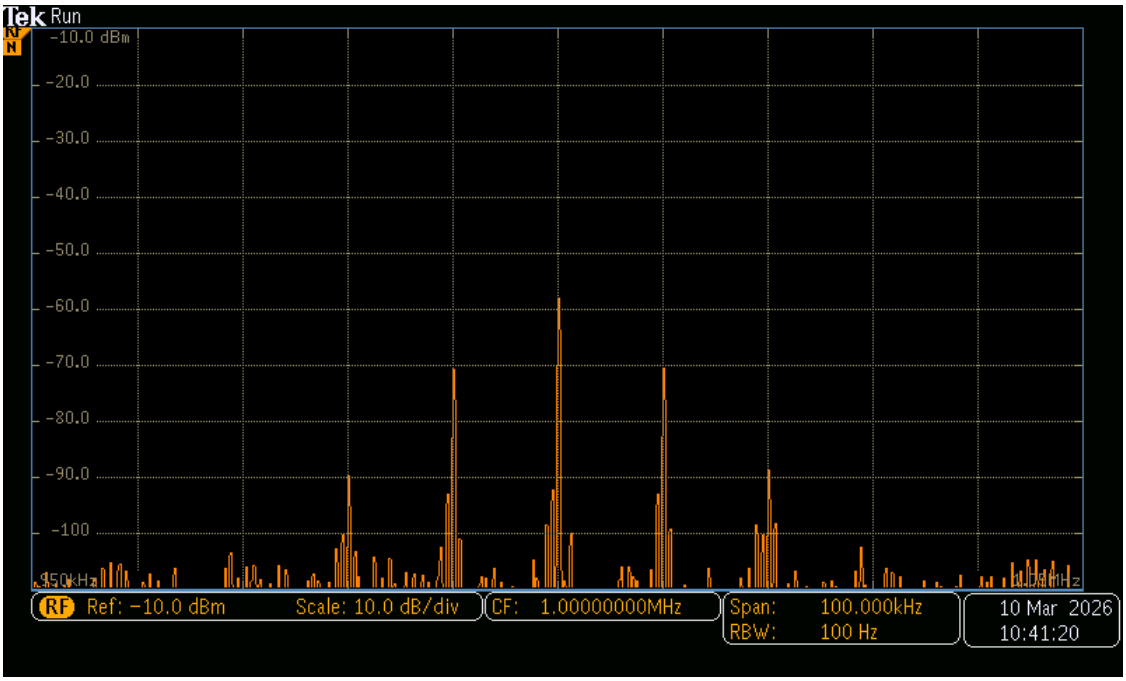

Figure 14: Frequency Spectrum with only the Carrier and First Sideband Pair

d) Only the carrier, 1st sideband pair, 2nd sideband pair and 3rd sideband pair are there

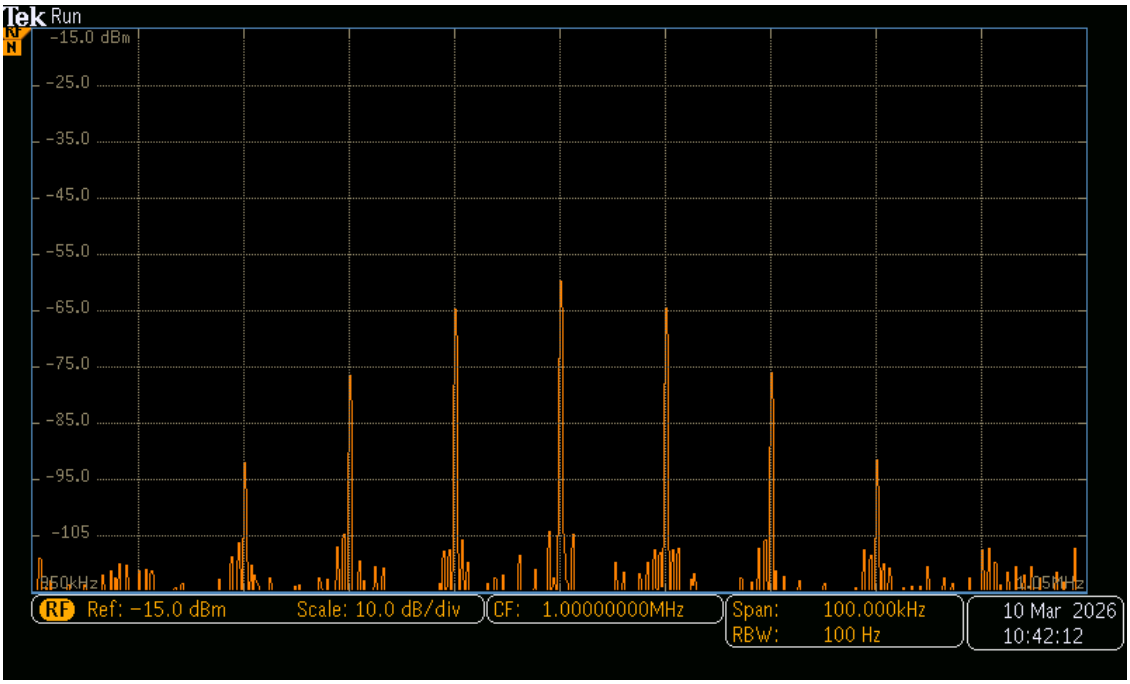

Figure 15: Frequency Spectrum with only the Carrier and First Sideband Pair

e) Suppressed carrier

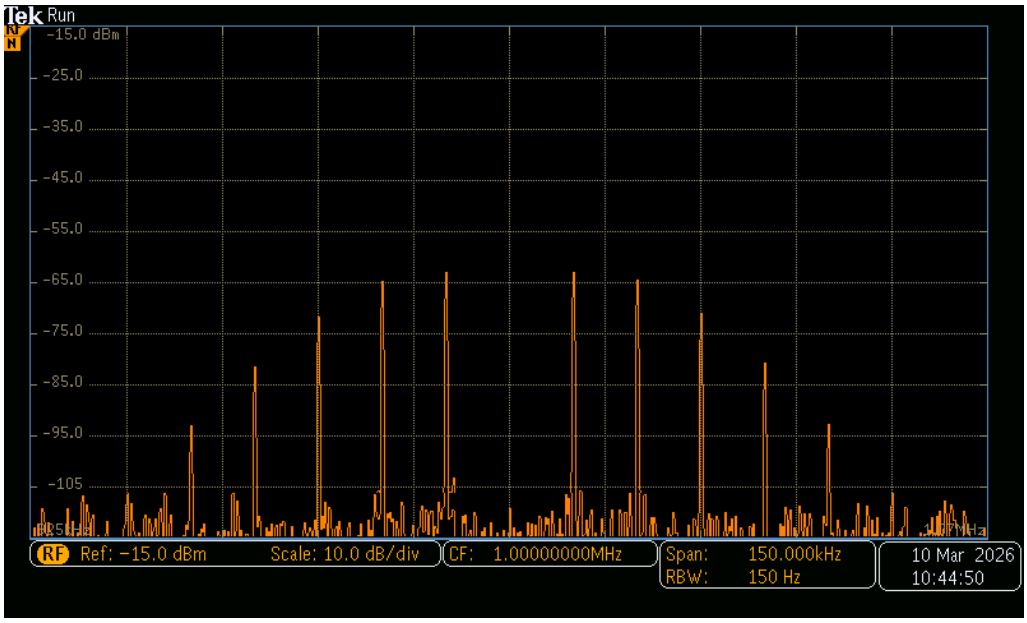

Figure 16: Frequency Spectrum suppressed carrier

f) Suppressed 1st sideband pair

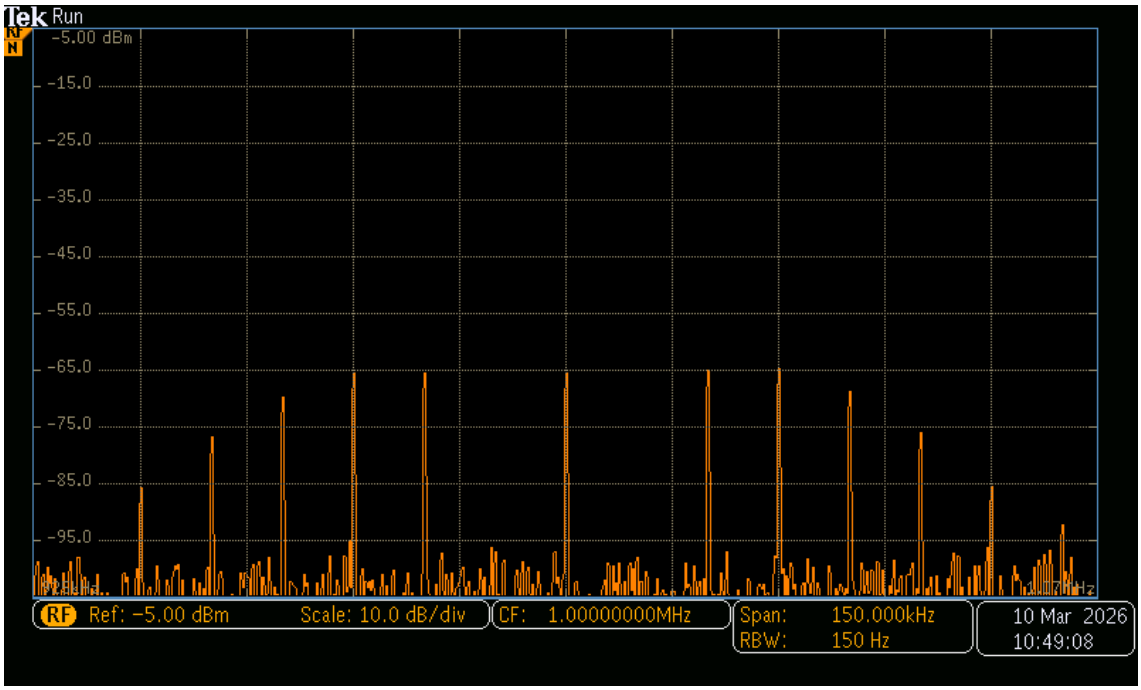

Figure 17: Frequency Spectrum of suppressed 1st sideband pair

a) Suppressed 2nd sideband pair

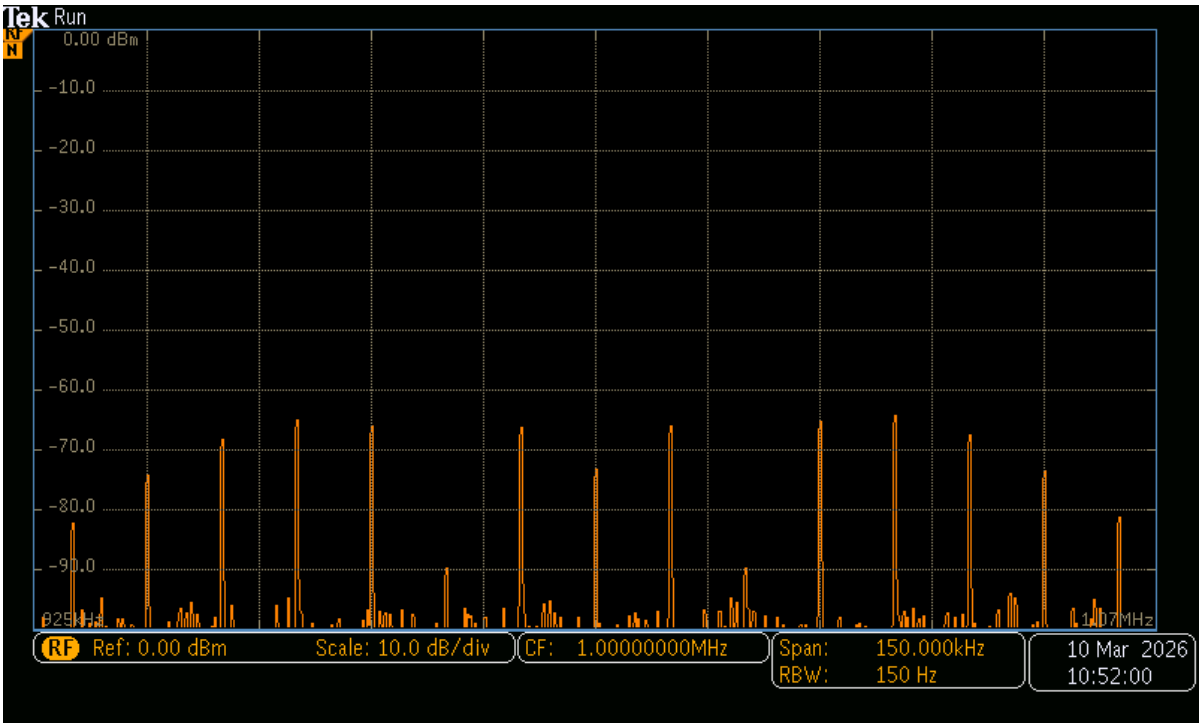

Figure 11: Frequency Spectrum when Modulation of suppressed 2nd side band pair

Part 4: Observe the effect of f_m on the frequency spectrum

f_m = 10 kHz

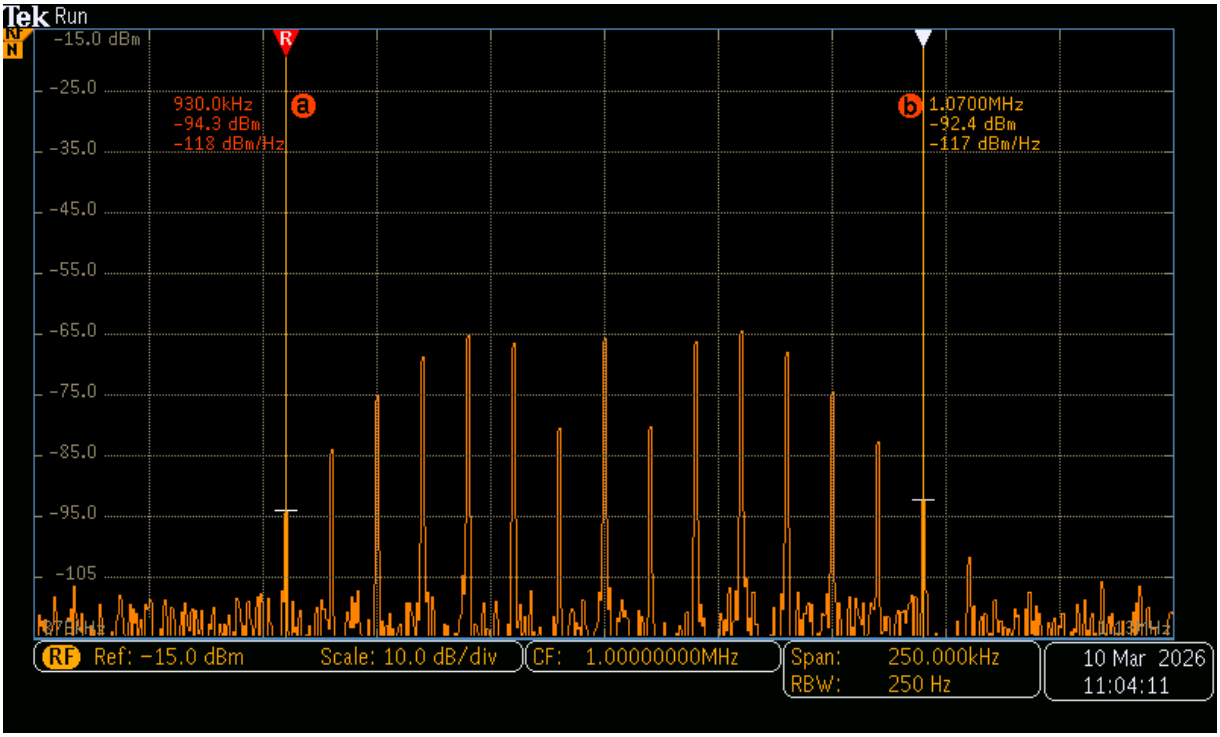

Figure 18: Frequency Spectrum with β = 4 and fm = 10 kHz

f_m = 15 kHz

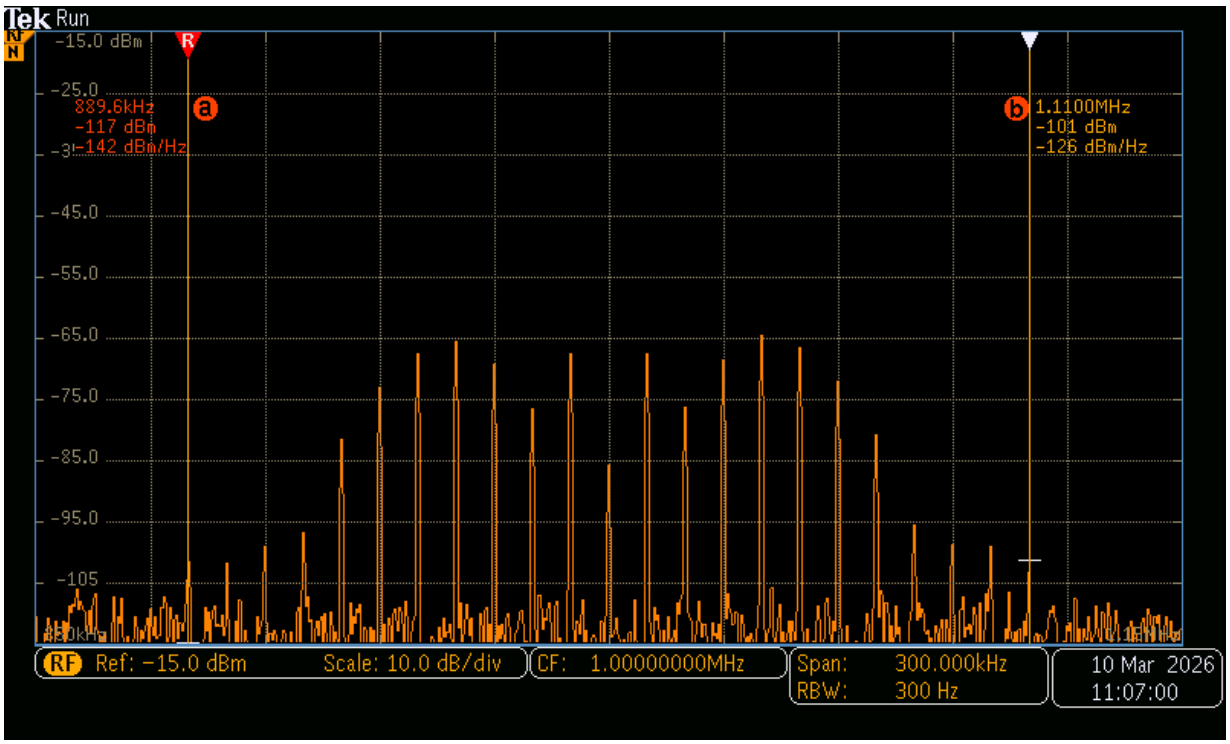

Figure 19: Frequency Spectrum with β = 4 and fm = 15 kHz

f_m = 20 kHz

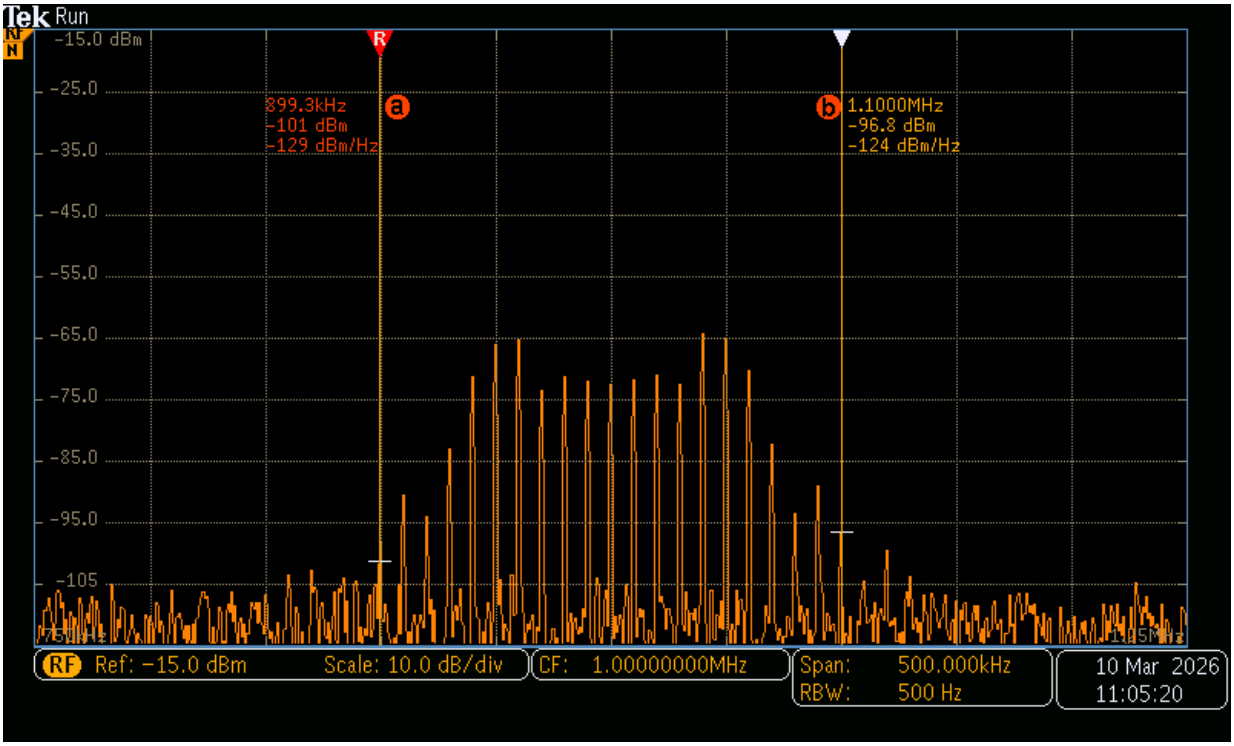

Figure 20: Frequency Spectrum with β = 4 and fm = 20 kHz

CALCUCLATIONS

Part 1: Identify the Sensitivity of the function generator
Gradient of the figure 3 gives the sensitivity

$$Sensitivity = \frac{1.0191 - 0.9718}{2 - (-3)}$$

$$Sensitivity = 0.00946\ MHzV^{-1} = 9.46\ kHzV^{-1}$$

Part 2: Generate different frequency-modulated signal which has different frequency spectrums for the same message signal

$$f_m = 10\ kHz$$

a) Carrier only

$$\beta = \frac{0\ kHz}{10\ kHz} = 0$$

b) Only the carrier and the 1st sideband pair are there

$$\beta = \frac{1.419\ kHz}{10\ kHz} = 0.1419$$

c) Only the carrier, 1st and 2nd sideband pairs are there

$$\beta = \frac{4.527kHz}{10kHz} = 0.4527$$

d) Only the carrier 1st, 2nd and 3rd sideband pairs are there

$$\beta = \frac{9.999kHz}{10kHz} = 0.9999$$

e) Suppressed carrier

$$\beta = \frac{24.062kHz}{10kHz} = 2.406$$

f) Suppressed 1st sideband pair

$$\beta = \frac{37.996\ kHz}{10kHz} = 3.7996$$

g) Suppressed 1st sideband pair

$$\beta = \frac{50.937\ kHz}{10\ kHz} = 5.0937$$

Table 2: Comparison of β values obtained using the Bessel Table and using the Observations of Frequency Spectrum

Frequency Spectrum Case	Calculated β using Observations	β values obtained using the Bessel Table
Carrier Only	0	0
Only the carrier and the 1st sideband pair are there	0.1419	0.25
Only the carrier, 1st and 2nd sideband pairs are there	0.4527	0.5
Only the carrier, 1st, 2nd and 3rd sideband pairs are there	0.9999	1.0
Suppressed Carrier	2.4062	2.4048
Suppressed 1st sideband pair	3.7996	3.8317
Suppressed 2nd sideband pair	5.0937	5.14

DISCUSSION

1. Explain why FM is better than AM in terms of resistance to noise

FM encodes information in the frequency of the carrier wave rather than its amplitude. Since noise mainly affects signal amplitude, it has little impact on the frequency-encoded information in an FM signal. In contrast, AM directly encodes the message in the amplitude, so any amplitude disturbance from noise corrupts the signal. FM receivers also use a limiter circuit that clips amplitude variations before demodulation, further rejecting noise. Additionally, the capture effect in FM means the receiver locks onto the strongest signal and suppresses weaker ones, including interference. FM also operates in the VHF band (88–108 MHz), which is generally less susceptible to atmospheric and man-made noise compared to the MF band used by AM. Together, these factors give FM a significantly better signal-to-noise ratio.

2. Describe two methods to estimate the occupied bandwidth of an FM signal

The first method is Carson's Rule, which gives a quick approximation of FM bandwidth as

$BW = 2(\Delta f + fm)$, where Δf is the peak frequency deviation and fm is the highest modulating frequency. It is simple to apply and accounts for the sidebands that carry most of the signal power, making it the most commonly used method in practice.

The second method uses the Bessel Function of the First Kind. An FM signal consists of a carrier and an infinite series of sidebands, with amplitudes given by $J_n(\beta)$. By identifying which sidebands have amplitudes above a defined threshold (1% of the unmodulated carrier), the significant spectral components can be counted, and the bandwidth is then taken as twice the frequency of the highest significant sideband. This method is more precise but requires Bessel function tables or computational tools.

3. Describe the key features of spectrum allocation used for typical FM radio broadcasting

FM radio broadcasting is allocated the VHF band from 88 MHz to 108 MHz. The spectrum is divided into channels with a standard spacing of 200 kHz, which is also the assigned bandwidth per channel. This is wide enough to carry high-fidelity stereo audio and additional data services such as RDS. The maximum permitted frequency deviation per station is ±75 kHz. Small guard bands are placed between adjacent channels to reduce co-channel interference. Transmitter power is also regulated to limit coverage to the intended service area and avoid interfering with stations in neighboring regions. These measures together allow efficient use of the spectrum while maintaining good audio quality.

4. Compare the performance of FM and AM on the frequency spectrum

AM signals have a narrow bandwidth of around 10 kHz, which allows many stations to coexist within the MF band (530–1700 kHz) but limits audio quality to speech-grade. FM signals occupy approximately 200 kHz per channel in the VHF band (88–108 MHz), enabling high-fidelity stereo audio and additional data. The wider bandwidth of FM means fewer channels fit in a given spectrum range, but the audio quality and noise immunity are far superior. In terms of propagation, AM signals can travel long distances, especially at night due to ionospheric reflection, making AM suitable for wide-area coverage. FM signals are limited to line-of-sight range, so coverage is more localized. Overall, AM trades audio quality for range and spectrum efficiency, while FM trades range for superior sound quality and noise resistance, making it the preferred choice for music broadcasting.

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